

Suggested Solutions for 2011 JC2 Preliminary Examination Paper 2

1. (a) K_w of water at 25°C is $1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$
 $\text{pOH} = (14 - 12.4) = 1.6$
 $[\text{OH}^-] = 10^{-1.6} = 0.0251 \text{ mol dm}^{-3}$
- (b) (i) Since $\text{H}^+ \equiv \text{OH}^-$
 $[\text{H}^+]_{\text{diluted}} = (0.0251 \times 25.0) / 24.50 = 0.0256 \text{ mol dm}^{-3}$
Volume of HCl required = $250 \times 0.0256 / 0.200 = 32.0 \text{ cm}^3$
- (ii) Using a 50.00 cm^3 burette, run/place 32.00 cm^3 of $0.200 \text{ mol dm}^{-3}$ HCl into a 250 cm^3 volumetric/standard flask.
Add de-ionised water and make up to the mark using deionised water.
Stopper the volumetric flask and shake well to obtain a homogeneous solution.
- (c) Procedure for Preparation of Saturated solution & Titration
- Using a 100 cm^3 measuring cylinder, place about 150 cm^3 of $0.100 \text{ mol dm}^{-3}$ CaCl_2 into a 250 cm^3 dry conical flask/beaker.
 - Using a spatula, add a large spatula of (solid) Ca(OH)_2 into the flask and shake. Continue to add more solid and swirl, until some Ca(OH)_2 remains undissolved to ensure a saturated solution is obtained.
 - Filter the saturated solution using a dry filter funnel and dry filter paper into dry conical flask/beaker to ensure the saturated solution is not diluted.
 - Pipette 25.0 cm^3 of saturated Ca(OH)_2 solution into another conical flask. Add 1-4 drops of methyl orange indicator. Titrate with the diluted HCl from the burette until the colour of the solution changes from yellow to orange.
 - Repeat the titration/step 4 to obtain two consistent titres/results OR 2 titres/results within 0.10 cm^3 in difference.
- (d) (i) $\text{Ca(OH)}_2 \equiv 2 \text{ HCl}$
Amount of Ca(OH)_2 in $25.0 \text{ cm}^3 = \frac{1}{2} (22.75 \times 10^{-3} \times 0.0245)$
 $= 2.79 \times 10^{-4} \text{ mol}$
Amount of Ca(OH)_2 in $1 \text{ dm}^3 = 2.79 \times 10^{-4} \times 1000 / 25.0$
 $= 0.0112 \text{ mol}$
Solubility of Ca(OH)_2 in $0.100 \text{ mol dm}^{-3}$ CaCl_2 at the laboratory temperature = $0.0112 \text{ mol dm}^{-3}$ OR $0.0111 \text{ mol dm}^{-3}$

$$\begin{aligned}
 \text{(ii)} \quad K_{sp} &= [\text{Ca}^{2+}][\text{OH}^-]^2 \\
 &= (0.1 + 0.0112)(2 \times 0.0112)^2 \\
 &= 5.58 \times 10^{-5} \text{ mol}^3 \text{ dm}^{-9}
 \end{aligned}$$

- (e) Solid $\text{Ca}(\text{OH})_2$ is corrosive/causes severe irritation/burns if it comes into contact with skin or eyes / harmful if swallowed.

Thus, to minimise risk, avoid contact with eyes / skin and clothes / wash hands thoroughly after handling and keep container closed.

OR

Dilute hydrochloric acid causes irritation to skin or eyes.

Wear gloves, goggles & laboratory coat to avoid direct contact.

2. (a) Entropy is the measure of disorder of the system.
- (b) $\Delta S_r^\circ(\text{CO})$ is negative as there is a decrease in disorder due to a decrease in number/amt of gas molecules/particles from 2 mol to 1 mol.

$$\begin{aligned}
 \text{(c)} \quad \text{(i)} \quad \Delta G_r^\circ &= \Delta H_r^\circ - T\Delta S_r^\circ \\
 +158 &= +157 - 298 \Delta S_r^\circ \\
 \Rightarrow \Delta S_r^\circ &= -3.36 \times 10^{-3} \text{ kJ K}^{-1} \text{ mol}^{-1} \\
 &= -3.36 \text{ J K}^{-1} \text{ mol}^{-1}
 \end{aligned}$$

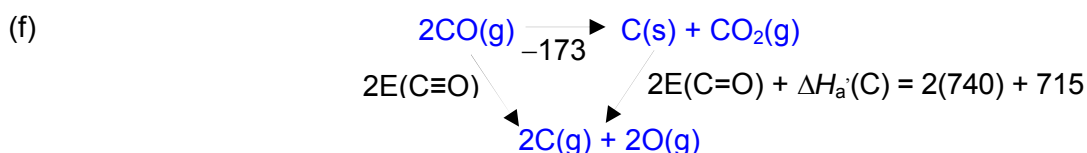
- (ii) The value is very close to zero (or very small/insignificant).
There is little/no change in disorder as 2 moles of solid produced 2 moles of solid.

- (d) Any value more negative than $-178 \text{ J K}^{-1} \text{ mol}^{-1}$

$$\text{(e)} \quad \Delta G_r^\circ = -RT \ln K_p$$

$$\ln K_p = -\left(\frac{-120 \times 1000}{8.31 \times 298}\right) = 48.5$$

$$K_p = e^{48.5} = 1.11 \times 10^{21} \text{ or } 1.16 \times 10^{21}$$



[1m] balanced energy cycle with state symbols and labels

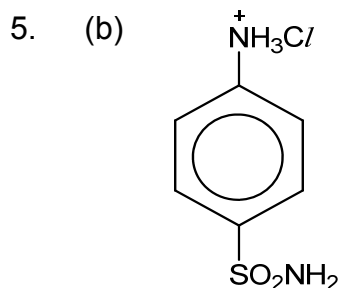
$$2\text{E(C}\equiv\text{O)} = (-173) + 2(740) + 715$$

$$\text{E(C}\equiv\text{O)} = +1011 \text{ kJ mol}^{-1} \text{ OR } +1010 \text{ kJ mol}^{-1}$$

- (g) The reaction is not spontaneous due to the high activation energy required to break the strong $\text{C}\equiv\text{O}$ in CO.

2. (h) Being a much stronger ligand than O_2 , CO ligand forms a stronger dative bond with the haemoglobin, hence displacing O_2 from the complex. This results in the inability of haemoglobin to transport oxygen to the cells and thus CO acts as a poison.
3. (a) +5
- (b) There is a decreasing trend as it is more difficult to remove H^+ from increasingly negatively charged species / anions.
- (c) conjugate acid: H_2PO_4 conjugate base: PO_4^{3-}
- (d) $K_b = \frac{1.00 \times 10^{-14}}{6.20 \times 10^{-8}} = 1.61 \times 10^{-7} \text{ mol dm}^{-3}$
- (e) (i) A buffer solution is one that resists changes in pH due to the addition of small amount of acid/ H^+ or base/ OH^- .
- (ii) $H_2PO_4^- + OH^- \rightarrow HPO_4^{2-} + H_2O$
- (iii) System: Acidic buffer
 $pH = -\lg(6.20 \times 10^{-8}) + \lg(1/2)$
 $= 6.91$
- (g) $[PO_4^{3-}] = (10/0 \times 0.0305) / 25.0 = 0.0122 \text{ mol dm}^{-3}$
- (h) 15 cm^3
- (i) (i) Bromocresol green
- (ii) blue to green

4. (a) Fe : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$
 Fe³⁺ : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$
- (b) Transition element complexes are coloured because of electron transition between d orbitals
 In a complex, the presence of ligands causes the 3d orbitals to split into 2 sets of different energies.
 The difference in energies (ΔE) between these two sets of 3d orbitals is so small such that radiation from the visible region of the electromagnetic spectrum is absorbed when an electron moves from a d orbital of lower energy to another partially filled / unfilled d orbital of higher energy.
 Hence, transition element complexes are coloured.
 The colour seen is the complement of the colours absorbed.
- (c) (i) Fe(NO₃)₃ (aq): [Fe(H₂O)₆]³⁺ (aq)
 E: [Fe(SCN)(H₂O)₅]²⁺ (aq)
- (ii) Fe(OH)₃ : Reddish-brown ppt
 Compound E: Blood-red solution
- (d) $E(\text{Fe}^{3+}/\text{Fe}^{2+}) = +0.77 \text{ V}$;
 $E(\text{Fe}(\text{CN})_6^{3-}/\text{Fe}(\text{CN})_6^{4-}) = +0.36 \text{ V}$
 $E(\text{Fe}(\text{OH})_3/\text{Fe}(\text{OH})_2) = -0.56 \text{ V}$
 $E(\text{Fe}^{3+}/\text{Fe}^{2+})$ is more positive than $E(\text{Fe}(\text{CN})_6^{3-}/\text{Fe}(\text{CN})_6^{4-})$ which is more positive than $E(\text{Fe}(\text{OH})_3/\text{Fe}(\text{OH})_2)$
 Oxidising Ability: Fe(OH)₃ , K₃Fe(CN)₆ , Fe(NO₃)₃
- (e) To 1 cm³ solution in a test tube, add 2 cm depth of aqueous sodium hydroxide. Warm/heat the mixture. Then add a piece of aluminium foil or zinc powder and warm if necessary. Test any gas evolved with damp red litmus paper.
5. (a) (i) 1. Concentrated HCl, Sn, heat
 2. NaOH/NaHCO₃/Na₂CO₃
- (ii) CH₃COCl
- (iii) electrophilic substitution



(c) It serves as a protecting group to protect the amine functional group.

(d) Sulfanilamide is insoluble in water due to the presence of bulky/large non polar benzene ring, which hinders the formation of effective hydrogen bonding with water molecules.

(e) (i) $\text{C}_6\text{H}_5\text{NH}_2 < (\text{CH}_3)_3\text{N} < \text{CH}_3\text{NH}_2$

(ii) $\text{C}_6\text{H}_5\text{NH}_2$ vs CH_3NH_2 & $(\text{CH}_3)_3\text{N}$:

For $\text{C}_6\text{H}_5\text{NH}_2$ or phenylamine, the lone pair of electrons on N atom is delocalised into the benzene ring, making it less available for protonation.

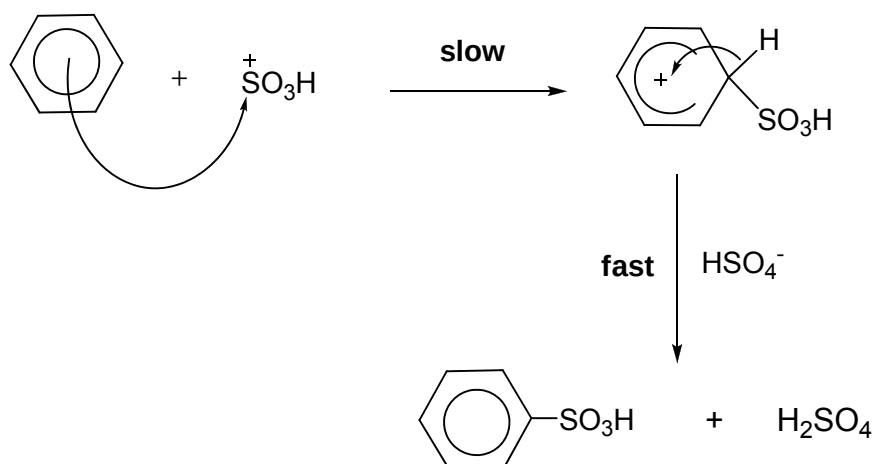
Thus, $\text{C}_6\text{H}_5\text{NH}_2$ or phenylamine is a weaker base than CH_3NH_2 & $(\text{CH}_3)_3\text{N}$.

CH_3NH_2 vs $(\text{CH}_3)_3\text{N}$:

$(\text{CH}_3)_3\text{N}$ or trimethylamine is a weaker base than CH_3NH_2 or methylamine due to the steric hindrance which makes the lone pair of electrons on N atom less available for protonation/ decreases the chances of protonation.

(f) (i) $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{SO}_3\text{H}^+ + \text{HSO}_4^-$

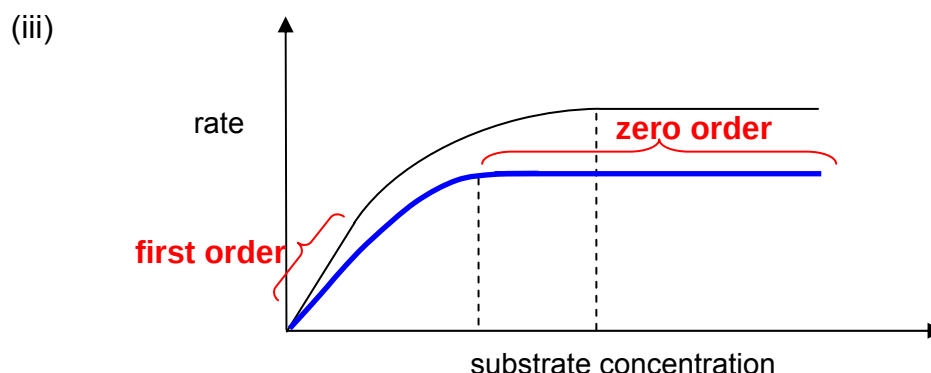
(ii)



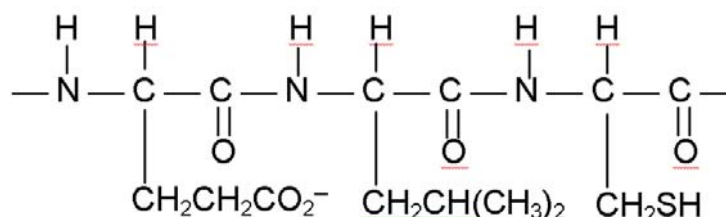
6. (a) Leu – Thr – Glu – Ile – Gly – Cys – Thr – Leu – Asn – Phe

(b) (i) When the substrate concentration is high, it shows a horizontal straight line (or flat) graph which implies that the rate of reaction is constant/independent of the substrate concentration.

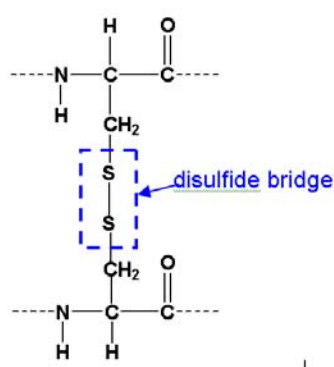
(ii) All active sites of all the enzyme/ HIV-1 Protease molecules are occupied with the substrate. No further increase in rate of reaction can take place because the enzyme is said to be saturated with the substrate.



(c) (i)



(ii) Side-chain interaction: Disulfide bridge



(iii) *Quaternary structure* of a protein is the spatial arrangement of two or more polypeptides chains held by R group interactions: ionic interactions, van der Waals' forces and hydrogen bonds to form a large, complex protein molecule.

6. (d) Extreme heat denatures/alters/destroys the shape/conformation of the HIV strand/protein because the weak interactions:

van der Waals' interactions in the tertiary structure and hydrogen bonds

in the secondary, tertiary and quaternary structures are disrupted/destroyed/broken.